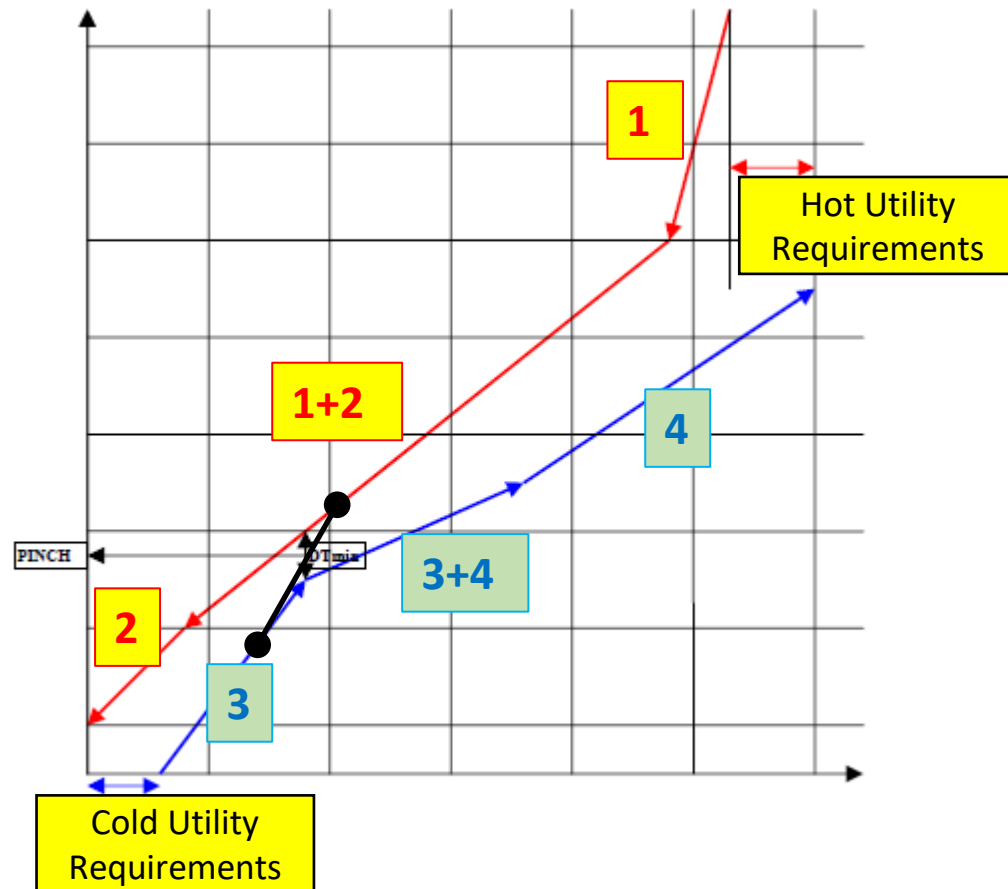


Synthesis of Heat Exchanger Networks (HEN)

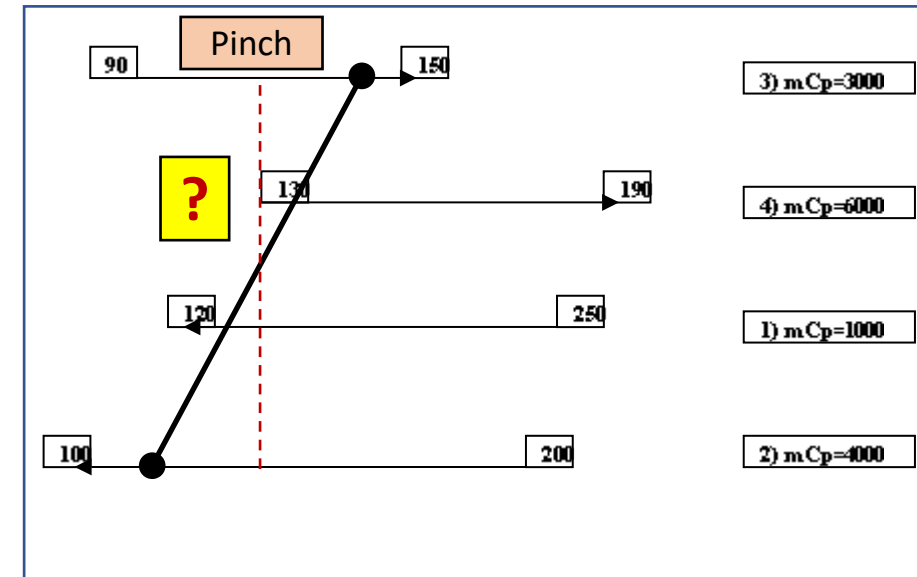
Basic Principle: Try to maintain "all" ΔT at their maximum values (minimum A) along the whole ΔH , using all feasible opportunities to save energy.

- **Heat transfer ... How?**
- Remember: The system is **NOT linear**
 - A ΔT of 5K combined with a ΔT of 15K leads to an average ΔT of 10K?

Basic Rules (1)



Stream	Condition	$mc_p \left(\frac{kJ}{h \cdot K} \right)$	$T_{in} (^{\circ}C)$	$T_{out} (^{\circ}C)$	$Q_{disp.} \left(\frac{kJ}{h} \right)$
1	HOT	1000	250	120	130 000
2	HOT	4000	200	100	400 000
3	COLD	3000	90	150	-180 000
4	COLD	6000	130	190	-360 000
					-10 000



Synthesis of Heat Exchanger Networks (HEN)

Basic Principle: Try to maintain "all" ΔT at **their maximum values** (minimum A) along the whole ΔH , using all feasible opportunities to save energy.

- **Heat transfer "in vertical"**
- Remember: The system is **NOT linear**
 - A ΔT of 5K combined with a ΔT of 15K leads to an average ΔT of 10K?

Advices:

1. Do NOT transfer heat across the Pinch
2. External heating ONLY above the Pinch
3. External cooling ONLY below the Pinch
4. Do NOT use thermal engines across the Pinch
5. Use ONLY Heat Pumps across the Pinch

Synthesis of Heat Exchanger Networks (HEN)

1.- Do not transfer heat through the Pinch. So:

- a) Design above the pinch
- b) Design below the pinch

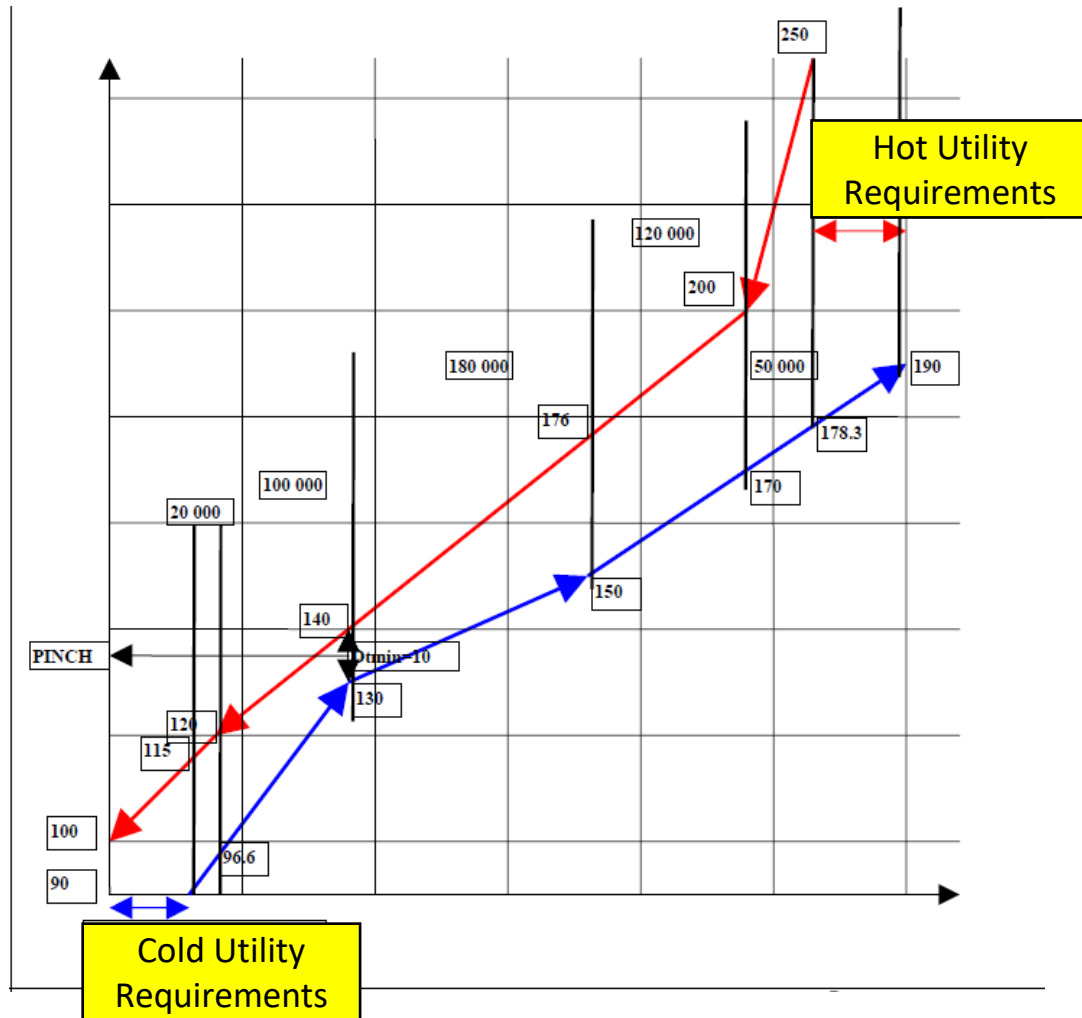
2.- Heuristics (= whenever it is feasible ... and specially close to the Pinch)

- a) Above the Pinch:
- b) Below the Pinch:
- c) Strictly maintain the intervals identified in the diagram

3.- **After 1+2:** consider other strategies:

- i. Try assignments “against Heuristic 2” (far away from the Pinch)
- ii. Try Stream splits
- iii. Combine Heat Exchangers
- iv. Analyse loops and circuits (connecting (or not) streams above and below the Pinch (against Heuristic 1))
- v. Analyse the Utilities specs (try to use more than one utility level)
- vi. Readjust DT_{min}
- vii. Integrate “mechanical” and “thermal” energy (CHP).

Basic Rules (2)



Stream	Condition	$mc_p \left(\frac{kJ}{h \cdot K} \right)$	$T_{in} (^{\circ}C)$	$T_{out} (^{\circ}C)$	$Q_{disp.} \left(\frac{kJ}{h} \right)$
1	HOT	1000	250	120	130 000
2	HOT	4000	200	100	400 000
3	COLD	3000	90	150	-180 000
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